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* Data structures (Multivalued variable)

=> Python provides several built-in data structures that help us organizing and manage data efficiently.

Different types of DS are:

1. List 2. Tuple 3. Set 4. Dict 5. Str

(1) List

=> It's is multivalued heterogenous data collectors.

- It is ordered, mutable and it will allow duplicates its represented by [].

Ex: l1 = [15, "kiran", 2004, 21.5, True]
l1
o/p [15, 'kiran', 2004, 21.5, True]

* l1.append(45)
l1
o/p [15, 'kiran', 2004, 21.5, True, 45]

* 11.insert (4, False)

11

O/P [15, 'Kiran', 2004, 21.5, False, True, 45]

* 11.pop()

11

O/P [15, 'Kiran', 2004, 21.5, False, True]

* 11.pop(3)

11

O/P [15, 'Kiran', 2004, False, True]

* 11.remove (False)

11

O/P [15, 'Kiran', 2004, True]

We displaying in the sequence

```
for i in 11:
    print(i)
```

O/P
15
Kiran
2004
True

we adding 2 list of the individual elements sum

```
l2 = [2, 5, 67, 89, 23, 12]
l3 = [76, 54, 23, 89, 23, 80]

sl = []
for i in range(0, 6, 1):
    s = l2[i] + l3[i]
    sl.append(s)
print(sl)
```

o/p [78, 59, 90, 178, 46, 92]

Ex: create. even, odd, prime number list from 1 to 20 numbers

```
en = []
on = []
pn = []
for i in range(1, 21, 1):
    if (i%2==0):
        en.append(i)
    else:
        on.append(i)
    if(i!=1):
```

```
        for j in range(2,i):
            if(i%j==0):
                break
        else:
            pn.append(i)
print(en)
print(on)
print(pn)
```

O/P
[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
[1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
[2, 3, 5, 7, 11, 13, 17, 19]

Ex1:
```
a=[1,2,3,4,5]
b=[4,5,6,7,8]
c=a
for i in b:
    if i not in a:
        c.append(i)
print(c)
```

O/P
[1, 2, 3, 4, 5, 6, 7, 8]

Ex1:
```
a=[1,2,3,4,5]
b=[4,5,6,7,8]
c=[]
for i in a:
```

```
        if i not in b:
            c.append(i)
for j in b:
    if j not in a:
        c.append(j)
print(c)
```

O/P [1, 2, 3, 6, 7, 8]

* slicing and list comphrension

slicing [start : end : inc/dec]

- default start is 0
- default end position is upto last position
- default 'inc/dec' is 1

Ex: list1 [15, "kiran", 87.9, True, 35, "Python", 88]

list1

O/P [15, 'kiran', 87.9, True, 35, 'python', 88]

- list1[0:3]

O/P [15, 'kiran', 87.9

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- Extract last 3 elements

```
list1[-3:]
```

O/P [35, 'Python', 88]

- Extract even indexed elements

```
list1 [1:7:2]
```

O/P [15, 87.9, 35, 88]

- Extract elements whoes index is multiple of 3

```
list1 [3::3]
```

O/P [True, 88]

- Reverse the list

```
list1 [::-1]
```

[88, 'Python', 35, True, 87.9, 'Kiran', 15]